Assignment-Regression Algorithm

1)Identify your problem statement

1.Machine Learning
2.Supervised Learning
3.Regression

2)Basic info about the dataset

Independent=1338 rows × 5 columns

Dependent= 1338 rows x 1 columns

3)pre-processing method

Ordinal method

5)

1.Multiple Linear Regression

r 2 = 0.789479034986701

2.Support Vector Regression

S.NO	Hyper Parameter	Linear (r value)	RBF (non linear) (r value)	Poly (r value)	Sigmoid (r value)
1	C=0.01	-0.0888313	-0.08964553	-0.08956828	-0.08956501
2	C=10	0.46246841	-0.03227322	0.038716222	0.039307143
3	C=100	0.62887928	0.320031783	0.617956962	0.52761035
4	C=1000	0.76493117	0.810206485	0.856648767	0.287470694
5	C=10000	0.74142301	0.877995240	0.859171507	-34.1515359
6	C=100000	0.74141889	0.872498445	0.857788112	-3465.9535

The regression use R2 value (rbf and hyper parameter(c=10000)) = 0.877995240

3.Decision Tree

s.no	criterion	splitter	R2	
1	squared_error	*best*	0.7063758813	
2	friedman_mse	*best*	0.6977768994	
3	absolute_error	*best*	0.6506332834	
4	poisson	*best*	0.7292183256	
5	squared_error	*random*	0.6962371450	
6	friedman_mse	*random*	0.7604962690	
7	absolute_error	*random*	0.7338694154	
8	poisson	*random*	0.7240738932	

The Decision tree regression r2 value is (freidman_mse,random) = 0.7604962690

4.Random Forest

S.No	criterion	n_estimators	R2 Value
1	squared_error	10	0.83303041340
2	squared_error	50	0.84983293150
3	squared_error	100	0.85383079130
4	absolute_error	10	0.83506355531
5	absolute_error	50	0.85266559935
6	absolute_error	100	0.85200936210
7	friedman_mse	10	0.83316626784
8	friedman_mse	50	0.85007161391
9	friedman_mse	100	0.85405189351
10	poisson	10	0.83139910401
11	poisson	50	0.849107595839
12	poisson	100	0.852633425889

The Random Forest regression r2 value is (friedman_mse, 100) =0.85405189351

5.Best Model is

Support Vector Machine r2 value is rbf and hyper parameter(c=10000)) = 0.877995240